Media Engineering and Technology Faculty
German University in Cairo





Market Mood in Two Languages: Bilingual Financial-News Sentiment for EGX30 Direction Forecasting

Bachelor Thesis

Author: Hussien Haitham Hussien Abdelmotaleb Hussien
Supervisors: Dr. Mervat AbuElkheir
Submission Date: 8 June 2026

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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor's degree

(ii) due acknowledgment has been made in the text to all other material used

Generative AI tools were used only for brainstorming, code boilerplate support, LaTeX drafting assistance, wording refinement, and checklist review. All scientific decisions, experimental design choices, data handling, results, citations, and final interpretations were verified by the author. No data, results, or citations were fabricated.

Hussien Haitham Hussien Abdelmotaleb Hussien
8 June 2026

Acknowledgments

This thesis would not exist without the people around me, and I want to take a moment to say that clearly.

To my family, my mother, my father, and my sister, thank you for everything: the support, the patience, and the belief even when I had none. This is as much yours as it is mine.

To my friends, thank you for being there throughout this journey. You know who you are and what it meant.

To Dr. Mervat AbuElkheir, you have been more than a supervisor to me. You are a mentor in the truest sense of the word, and the way you guided this work and believed in me throughout made a real difference. Thank you.

To my supervising team at Robin, Mostafa El Far, Logine El Kelany, and Abdelrahman Hassan, thank you for the guidance, the feedback, and for taking this project seriously. Working with you made this work better.

A special thank you to Omar Kamal, who did the interviews at the beginning and believed in what this could become before it became anything. That belief mattered more than I can properly express.

Thank you to Basma Rady, Managing Director of Robin, for creating an environment where a project like this was even possible.

To the entire Robin team, thank you. The collaboration, the openness, and the warmth throughout this project meant a great deal. I am grateful to have worked with all of you.

Finally, thank you to the German University in Cairo for five years that shaped how I think, work, and approach problems. This thesis is the product of everything that happened here.

Abstract

This thesis studies whether bilingual Arabic-English financial news contains short-horizon directional information for the Egyptian Exchange EGX30 index. The setting is an emerging-market information environment where firm-related news appears in both Arabic and English, and where using the current EGX30 list retrospectively would create survivorship bias.

The study reconstructs a point-in-time (PIT) EGX30 constituent universe from 21 April 2021 to 20 April 2026. The final registry contains 12 continuous membership intervals and a historical union of 57 companies. Arabic and English Mubasher articles linked to those historical constituents are collected, normalized, filtered by active membership, deduplicated across company pages, and scored with language-specific sentiment models. English articles are scored with FinBERT, while the Arabic route compares AraBERT and CAMeLBERT.

The primary forecasting design is intentionally sentiment-only. EGX30 historical closes are used only to construct 1-, 3-, and 5-trading-day direction labels, and no closes, returns, Open, High, Low, Close, Volume (OHLCV) fields, technical indicators, or other price-derived variables enter the main model matrix. Daily company sentiment is aggregated into an equal-weight market-mood signal, shifted before trading-date alignment, and evaluated with strict expanding-window walk-forward validation.

The strongest bilingual test row by balanced accuracy is FinBERT + CAMeLBERT at the 1-trading-day horizon, with accuracy 0.621, balanced accuracy 0.545, F1 0.737, and Matthews Correlation Coefficient (MCC) 0.115. Baseline and class-imbalance checks show that raw accuracy is partly explained by majority-class behavior, and only 8 of 18 sentiment-model test rows beat the training-majority baseline on balanced accuracy. A supplementary XGBoost integration experiment adds the same lagged sentiment features to an EGX30 historical OHLCV model; accuracy improves from 0.739 to 0.802 and balanced accuracy improves from 0.757 to 0.793. A validity audit passes target recomputation, lag alignment, point-in-time membership, cross-company deduplication, walk-forward history, and metric-parity checks.

The results provide promising but limited evidence of weak directional signal in bilingual EGX30 market mood. The contribution is a reproducible and conservative protocol for testing bilingual market mood in an emerging market, rather than a deployable trading system.

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Chapter 1

Introduction

Financial markets allocate capital, transfer risk, and react continuously to public information. When listed firms release news or when financial media reports new developments, investors update expectations about value and risk. For analysts and research desks, the practical question is not simply whether prices move. It is whether public information can be processed in a disciplined way before its relation to future market direction disappears.

Short-horizon forecasting is difficult because market reactions are noisy and unstable. Public information can be absorbed quickly, and the same type of news may have different effects across market regimes. The efficient-market literature explains why persistent predictability is hard to find, while empirical forecasting studies show that weak and conditional patterns may still exist under carefully defined targets and validation protocols [1, 2, 3].

Financial news is relevant because it contains semantic information that price histories do not directly encode. A price series records that a movement occurred, but it does not explain whether the movement followed earnings news, financing plans, governance changes, regulation, or contracts. Textual sentiment is therefore useful as a research signal because it tests whether the tone and relevance of public news contain incremental information beyond numerical market history [4, 5].

Most sentiment-forecasting work focuses on developed markets, English text, social media, or models that combine text with price-derived predictors. Those settings are useful but do not directly answer the Egyptian-market question. The Egyptian Exchange is a bilingual Arabic-English information environment, and the EGX30 index provides a broad market target that is more appropriate for market-mood aggregation than a small set of selected equities.

This thesis studies a narrow problem: whether bilingual financial-news sentiment from point-in-time EGX30 constituents contains short-horizon directional information for the EGX30 index. The feature matrix excludes historical prices, EGX30 closes are used only to build labels, and balanced accuracy is treated as more important than raw accuracy because the direction labels are imbalanced.

1.1 Problem Statement

There is limited reproducible evidence on whether Arabic-English financial-news sentiment from the historical EGX30 constituent universe contains short-horizon directional

information for the EGX30 index. Existing work often uses English-only data, current constituent lists, price-derived predictors, or evaluation protocols that are difficult to audit. This creates a gap for an Egyptian-market study that combines point-in-time market coverage, bilingual sentiment processing, sentiment-only predictors, and strict walk-forward validation.

1.2 Motivation

The main motivation is methodological clarity. If sentiment is mixed immediately with prices, returns, volume, or technical indicators, it becomes difficult to tell whether text itself contributes useful information. A sentiment-only benchmark isolates the textual signal first. Later price-plus-sentiment models can be evaluated more fairly only after this baseline is understood.

The second motivation is local relevance. EGX-related news is not purely English and not purely Arabic. A method that ignores either language loses part of the information environment, while mixing both languages without routing can introduce avoidable noise. Comparing Arabic encoder choices is part of the core study, not a separate side experiment.

The third motivation is reproducibility. The EGX30 changes over time, so using the current index composition to collect past news would bias the study toward firms that survived into the present list. Reconstructing the point-in-time constituent universe makes the corpus and the market target consistent.

1.3 Research Questions

RQ1. Does bilingual financial-news sentiment from point-in-time EGX30 constituents contain short-horizon directional information for the EGX30 index?

RQ2. How do Arabic sentiment encoder choice and forecast horizon affect sentiment-only EGX30 direction classification performance?

RQ3. Do any observed results remain meaningful under leakage-safe walk-forward validation, class-imbalance diagnostics, and stronger baseline comparisons?

1.4 Objectives

Objective 1. Reconstruct a point-in-time EGX30 constituent universe for the study period.

Objective 2. Collect, normalize, and score Arabic and English financial-news articles linked to historical EGX30 constituents.

Objective 3. Construct sentiment-only daily market-mood features while excluding all price-derived predictors from the model matrix.

Objective 4. Define 1-, 3-, and 5-trading-day EGX30 direction labels using historical closes only for target construction.

Objective 5. Evaluate sentiment-only forecasting performance under strict expanding-window walk-forward validation.

Objective 6. Audit coverage, leakage, class imbalance, baseline performance, and robustness so that the results are interpreted conservatively.

1.5 Thesis Outline

The thesis keeps the existing six-chapter structure of the template. Chapter 2 introduces financial forecasting, directional labels, sentiment analysis, bilingual NLP, and leakage-safe evaluation. Chapter 3 reviews prior work and narrows the research gap. Chapter 4 presents the proposed point-in-time bilingual sentiment pipeline. Chapter 5 reports dataset coverage, baseline comparisons, main results, robustness checks, and validity audits. Chapter 6 concludes the thesis and outlines focused future work.

1.6 Chapter Summary

This chapter established the central story of the thesis. The problem is not to build a deployable trading system, but to test whether bilingual market mood contains weak directional information for EGX30 index classification. The research questions and objectives therefore focus on point-in-time coverage, sentiment-only feature construction, and strict evaluation. The next chapter introduces the background concepts needed to justify those design choices.

Chapter 2

Background

This chapter introduces the ideas behind the thesis design. It starts with financial forecasting and limited predictability, then narrows to directional labels, class imbalance, financial-news sentiment, domain-specific NLP, Arabic and bilingual text processing, and leakage-safe evaluation. The goal is to show why the method in Chapter 4 avoids price-derived features and uses strict chronological testing.

2.1 Financial Forecasting and Limited Predictability

Financial forecasting sits between economics, statistics, and machine learning. Prices react to public information, liquidity, expectations, and strategic trading. The efficient-market view argues that public information is rapidly incorporated into prices, which makes persistent predictability difficult [1]. Later work questions the strongest random-walk interpretation, but still supports the view that predictable structure is weak and unstable [2].

This matters because a short-horizon forecasting result should not be accepted only because it has attractive raw accuracy. A useful study must define the target carefully, avoid future information, and compare against simple baselines. In this thesis, the target is not a future price level. It is the direction of the EGX30 index over short trading-day horizons.

The distinction is important. Predicting a price level asks the model to estimate magnitude and direction at the same time. Direction classification asks a narrower question: whether the future index return is positive or negative. That narrower target fits the goal of this thesis because the study tests whether news sentiment has directional information, not whether it can estimate the size of the next market move.

2.2 Directional Labels and Input Modalities

Forecasting studies use several targets, including price level, return magnitude, return sign, and binary direction. Direction classification is common because it maps naturally to whether the market moved up or down over a horizon. It also loses information about

magnitude, so direction results must be interpreted as classification evidence rather than as return forecasts [3].

Input modalities also differ. Price-based models use historical closes, returns, volumes, and technical indicators. Text-aware models use news, disclosures, social media, or analyst commentary. This thesis excludes price-derived predictors from the model matrix so that the news signal can be tested before any future multimodal extension.

2.3 Class Imbalance and Evaluation Sensitivity

Directional labels are often imbalanced. In an upward-trending period, always predicting an Up move can produce high raw accuracy while failing to discriminate between market states. Balanced accuracy is central here because it averages performance over both classes and is less forgiving when a classifier collapses toward the majority class.

F1, precision, recall, Matthews Correlation Coefficient (MCC), confusion-matrix counts, actual Up share, and predicted Up share provide additional context. These metrics do not replace balanced accuracy, but they help explain whether a model is learning a sentiment relation or simply following the dominant class. Stronger naive baselines are therefore part of the evaluation design rather than an optional appendix.

This issue is not only statistical. It changes the story a result can support. A model that mostly predicts Up may look useful in a rising market, but it has not necessarily learned when sentiment separates positive and negative future sessions. The thesis therefore treats class balance as part of the research question, not as a small reporting detail.

2.4 Financial News and Sentiment

Financial news can describe information that historical prices do not encode directly. Prior work shows that media tone, message-board language, and financial dictionaries can relate to market outcomes, but the usefulness of the signal depends on source quality, entity relevance, and aggregation [4, 6, 7, 5].

Financial news is usually cleaner than social media because articles are longer, more structured, and more often tied to identifiable companies. It is still imperfect. A company-page article may be weakly related to the firm, repeated across pages, or published outside trading hours. These issues motivate relevance weighting, deduplication, and lagged alignment in the methodology.

2.5 Domain-Specific Financial NLP

General sentiment models can misread financial language. Words such as “liability”, “capital”, or “debt” may have domain-specific meanings that differ from everyday sentiment. FinBERT addresses this issue for English financial text and is therefore a suitable English sentiment encoder for this thesis [8].

The choice of sentiment model matters because sentence-level errors accumulate into daily market-mood features. A weak sentiment classifier can create noisy aggregate signals even if the forecasting model is correctly validated. For this reason, the thesis treats sentiment scoring as part of the research method rather than as a black-box preprocessing step.

Domain fit matters more in finance than in many general text tasks. A word that sounds negative in everyday language may be neutral in a balance-sheet context, and a short earnings headline may carry information that a generic sentiment model treats as ordinary business language. This is why the English side uses a financial model instead of a general sentiment classifier.

2.6 Arabic and Bilingual Financial NLP

Arabic financial NLP adds linguistic complexity. Arabic is morphologically rich and appears across formal, dialectal, and domain-specific variants. Arabic-specific transformer models such as AraBERT, ARBERT, MARBERT, and CAMeLBERT show that language and variant proximity can matter for downstream NLP performance [9, 10, 11].

The EGX setting is bilingual. English articles and Arabic articles are not simply translations of the same information stream. They may differ in timing, wording, and coverage. This justifies a language-routing design: English articles are scored with an English financial model, while Arabic articles are scored with Arabic encoders and compared in the final evaluation.

This also means that bilingual processing is not just a translation problem. Translating all Arabic articles into English would add another source of error and could remove local financial wording. Routing each language through a more suitable model keeps the original text closer to its market context.

2.7 Chronological Evaluation and Leakage

Time-series forecasting is vulnerable to leakage. Future information can enter through same-day article alignment, overlapping target windows, hyperparameter tuning on test data, or refitting with labels that would not yet be known at prediction time. Text data makes this harder because articles have timestamps while targets are defined on exchange trading calendars [12, 13].

Chronological validation is therefore part of the problem definition. The thesis uses train, validation, and test regions in time order, adds embargo buffers, tunes thresholds on validation predictions only, and generates validation and test predictions through walk-forward histories. These choices make the reported results harder to inflate and easier to audit.

2.8 Chapter Summary

This chapter established the background logic for the thesis. Short-horizon market direction classification is difficult, sentiment is potentially useful but noisy, and bilingual EGX

news requires language-aware processing. Evaluation must prioritize balanced accuracy and leakage control because raw accuracy can be misleading. The next chapter reviews related work and identifies the specific gap addressed by this thesis.

Chapter 3

Related Work

This chapter reviews work that motivates the thesis and narrows the research gap. It does not treat every prior experiment as a model to copy. Instead, it asks what each line of work teaches about source choice, language, target definition, baselines, and leakage-safe evaluation.

3.1 Text-Aware Financial Forecasting

Early stock-forecasting systems relied mainly on prices, returns, volumes, and technical indicators. Text-aware forecasting expanded the feature space by adding news, social media, or event representations. AZFinText showed that breaking financial news can support short-term directional prediction when entity linkage and timing are controlled [14]. Later work improved text representation through context-aware features, event embeddings, attention models, and deep fusion architectures [15, 16, 17, 18].

The lesson for this thesis is not that more complex models are always better. The lesson is that text must be linked to the correct market entity and aligned to the correct prediction horizon. A bilingual EGX30 study therefore needs a defensible corpus construction step before any model result can be interpreted.

3.2 Social Media and Market Mood

Social media is attractive because it is timely and large in volume. Studies of Twitter mood, message-board posts, and search signals report that aggregate public attention can relate to market behavior [19, 20, 21]. Arabic social-media work also suggests that Arabic investor discussion may contain information relevant to Arab market indices, including EGX30 [22].

These studies are useful as evidence that public language can matter, but they do not define the main scope of this thesis. Social media has access, noise, entity-linking, and redistribution constraints. For the final bachelor thesis, social media is therefore treated as future work, while the core study uses auditable financial-news pages.

3.3 Financial News and Stock Prediction

Financial-news studies are closer to the present problem because news articles are more structured and more likely to refer to identifiable firms. Large-scale U.S. studies combine company news with prices and deep models [23]. Other studies compare recurrent models with and without news sentiment, or fuse sentiment with chart representations [13, 24]. Index-level models such as SA-DLSTM also show how online sentiment and market variables can be integrated for broad-market movement prediction [25].

These studies show that sentiment can be useful, but their settings differ from this thesis. Many use developed markets, English text, social-media sentiment, or price-derived predictors. The final EGX30 study instead asks a narrower question: whether bilingual constituent-linked financial-news sentiment alone contains directional information for an Egyptian index target.

3.4 Domain-Specific and Arabic Sentiment Models

The English side of financial sentiment has strong domain-specific support. FinBERT was developed for financial text and improves over generic language models on financial sentiment tasks [8]. This makes it a suitable English route for the thesis.

The Arabic side is less settled for finance-specific sentiment. AraBERT, ARBERT, MARBERT, and CAMeLBERT provide strong Arabic representations, but their benchmark success does not automatically establish forecasting value in EGX news [9, 10, 11]. This motivates comparing Arabic encoder routes inside the same sentiment-only EGX30 protocol.

3.5 Evaluation Practice and Baselines

A recurring limitation in sentiment forecasting is inconsistent evaluation. Some studies report regression error, others report accuracy, and fewer report balanced accuracy or class-distribution diagnostics. Horizon-comparative evidence shows that sentiment effects are conditional on target horizon and feature setup rather than universal [12].

For this reason, the present thesis treats baseline comparison as part of the main evidence. Always-Up, always-Down, training-majority, rolling-majority, article-count-only, and shuffled-sentiment baselines test whether the sentiment model is doing more than following label skew or news volume. Results are interpreted primarily through balanced accuracy, not raw accuracy.

3.6 Related-Work Comparison Matrix

Table 3.1 summarizes representative studies using the dimensions most relevant to this thesis. The purpose is not to rank the studies, but to show which methodological gaps remain for a bilingual Egyptian-market study.

Table 3.1: Comparison matrix for representative sentiment-based forecasting studies.

Study	Market	Language	Source	Sentiment / representation	Target	Evaluation	Relevance / limitation
Bollen et al. [19]	DJIA	En	Twitter	Mood	Index dir.	Acc.	Aggregate mood; social-media/developed-market setting
Nguyen et al. [20]	18 stocks	En	Boards	Topic sentiment	Stock dir.	Acc.	Entity filtering; not bilingual financial news
Mohan et al. [23]	S&P 500	En	News	Deep fusion	Stock dir.	Model cmp.	Large panel; developed-market and price-fusion focused
Dahal et al. [13]	Dev. stocks	En	News + market	Sentiment	Return reg.	RMSE, MAPE	Regression focus; not sentiment-only EGX30 direction
Huang and Ho [24]	U.S. stocks	En	Social + charts	Fusion	Stock dir.	Acc.	Uses price-derived visual inputs
Zhao et al. [25]	Hang Seng	Chinese	Opinion + market	Deep sentiment	Index dir.	Baselines	Index target; not Arabic-English news
Liapis et al. [12]	Benchmark	En	Sentiment + prices	Many setups	Horizons	Broad cmp.	Supports horizon-specific caution
Arabic encoder studies [9, 11]	Arabic NLP	Ar	Benchmarks	Encoders	NLP tasks	F1	Justifies routing; not an EGX30 forecast test

The table shows that prior work rarely combines all constraints required here. Developed-market studies have richer coverage but weaker relevance to EGX. Social-media studies are timely but noisy and difficult to audit. Arabic NLP studies justify language-specific encoders but do not test EGX30 direction classification.

The comparison also shows why the final study should not be framed as a simple application of an existing model. If the market setting is changed to Egypt, the language setting becomes bilingual. If the target is changed to EGX30, the news corpus must follow historical index membership. If price-derived predictors are removed, the model has less information but the interpretation becomes cleaner. If walk-forward validation and class-imbalance baselines are enforced, some attractive raw-accuracy results become much weaker.

Taken together, these constraints define the thesis contribution. The study combines an Egyptian index target, Arabic-English news, point-in-time constituent coverage, sentiment-only predictors, chronological validation, and baseline checks in one reproducible setup. Prior work covers many of these pieces separately, but rarely all at once.

3.7 Research Gap

The remaining gap is geographic and methodological. Most prior work does not simultaneously address the Egyptian market, bilingual Arabic-English financial news, point-in-time index constituent coverage, sentiment-only predictors, strict walk-forward validation, and class-imbalance-aware baselines.

This thesis addresses that gap by treating the evaluation setup itself as the contribution. The aim is not to claim strong market predictability. The aim is to provide a reproducible test of whether bilingual EGX30 market mood contains weak directional signal under careful validation.

3.8 Chapter Summary

This chapter reviewed prior work and narrowed the research gap. The literature supports the relevance of financial text, domain-specific sentiment models, and language-aware processing, but it also warns that sentiment effects are conditional and sensitive to evaluation design. The next chapter turns these lessons into the thesis methodology: a point-in-time, bilingual, sentiment-only EGX30 forecasting pipeline.

Chapter 4

Methodology

This chapter explains the method used in the thesis. The motivation is to isolate bilingual financial-news sentiment before adding any price-based information, so the method starts from historical EGX30 membership, collects bilingual company-linked news, routes text to suitable sentiment models, builds daily market mood, constructs EGX30 direction labels, and evaluates the result with walk-forward validation.

4.1 Overall Pipeline Architecture

The method isolates the news signal before adding any price-based information. Historical EGX30 closes are used only after feature construction, and only to define future direction labels. Figure 4.1 shows the end-to-end flow.

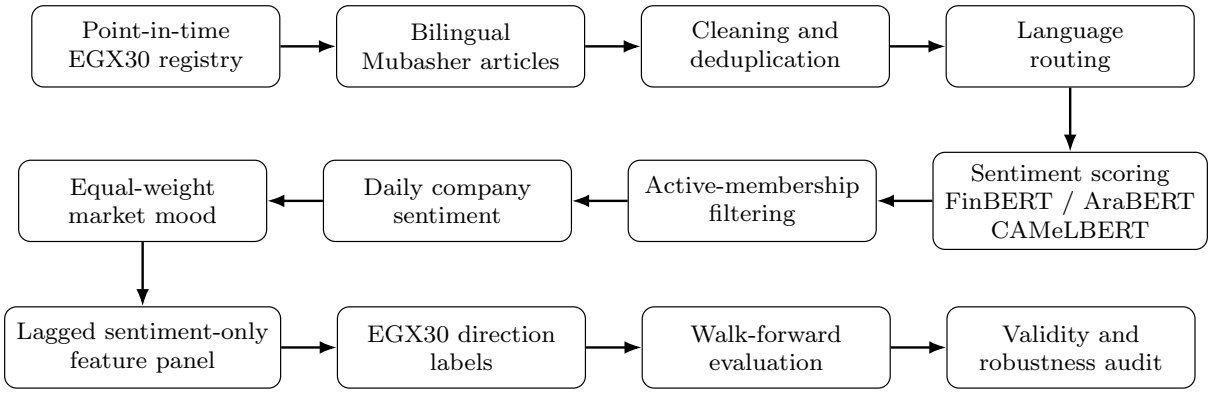


Figure 4.1: Overall pipeline architecture for the EGX30 sentiment-only study.

The ordering in Figure 4.1 is important. Articles are filtered against the point-in-time universe before aggregation, and sentiment features are lagged before they are aligned to trading dates. This prevents same-day information and future labels from entering the prediction procedure.

Algorithm 4.1 summarizes the construction of the final sentiment feature panel. The important constraint is that prices enter only after feature construction, and only to define the direction labels.

Algorithm 4.1: Sentiment Feature Panel Construction

Input: point-in-time EGX30 registry, article table, sentiment scores, trading calendar, EGX30 close series for labels only, and forecast horizons $H = \{1, 3, 5\}$.
Output: lagged sentiment-only feature panels and direction labels.

1. Normalize article timestamps to Cairo local dates.
2. Remove articles outside the study period.
3. Deduplicate repeated article records.
4. Keep an article only when its linked company is active in EGX30 on the article date.
5. Route text by language.
6. Score English text with FinBERT and Arabic text with AraBERT or CAMeLBERT.
7. Aggregate scored articles to company-day sentiment.
8. Aggregate active company-day sentiment to equal-weight daily market mood.
9. Build sentiment-side coverage and count features.
10. Shift all sentiment features by at least one calendar day.
11. Align shifted features to trading dates.
12. Construct EGX30 direction labels from future closes only.
13. Remove all price-derived fields from the model matrix.
14. Return the feature panel for walk-forward evaluation.

The output of Algorithm 4.1 is the only input used by the forecasting models. Because all sentiment features are shifted before trading-date alignment, the model cannot use same-day or future information.

4.2 Point-in-Time EGX30 Universe

Using the current EGX30 list for past dates would introduce survivorship bias. The study therefore reconstructs the active constituent set from 21 April 2021 through 20 April 2026. The registry contains 12 continuous intervals and 57 historical company identities.

Table 4.1 reports the intervals used by the final study. Two non-30 states are retained because they correspond to documented corporate-action transition periods. The registry validator checks continuity, rejects gaps and overlaps, and permits only those documented exceptions.

4.3 Bilingual Article Collection and Preprocessing

The news corpus is built from Arabic and English Mubasher company pages linked to the historical EGX30 constituent union. Each article is normalized to a Cairo local publication date. Rows outside the study period are removed, duplicate article records are collapsed, and article text is cleaned before sentiment scoring.

Table 4.1: Point-in-time EGX30 registry intervals and active company counts.

From	To	Active companies
2021-04-21	2021-07-31	31
2021-08-01	2021-10-12	30
2021-10-13	2022-01-31	32
2022-02-01	2022-07-31	30
2022-08-01	2023-01-31	30
2023-02-01	2023-07-31	30
2023-08-01	2024-01-31	30
2024-02-01	2024-07-31	30
2024-08-01	2025-02-01	30
2025-02-02	2025-08-02	30
2025-08-03	2026-01-31	30
2026-02-01	2026-04-20	30

Preprocessing also preserves audit fields such as source language, detected language, article identifier, URL, company ticker, and publication timestamp. These fields make the final corpus reproducible and support later checks for membership validity and cross-company duplication.

4.4 Sentiment Scoring

The scoring step routes English articles to FinBERT and Arabic articles to Arabic transformer encoders. The main Arabic route uses AraBERT, and the comparison route uses CAMeLBERT. The English route is kept fixed so that differences between the two bilingual sentiment sources mainly reflect the Arabic encoder choice.

For article i, let s_i be the continuous sentiment score and w_i be the article weight from relevance and confidence. Equation 4.1 defines the weighted article contribution:

$$q_i = s_i w_i. \tag{4.1}$$

This contribution is still a text-derived quantity. It does not use prices, returns, or trading volume.

4.5 Market-Mood Feature Construction

Article sentiment is first aggregated at the company-day level. For company c on calendar day d, Equation 4.2 defines the daily company signal:

$$S_{c,d} = \frac{\sum_{i \in A_{c,d}} q_i}{\sum_{i \in A_{c,d}} w_i}, \tag{4.2}$$

where $A_{c,d}$ is the set of scored articles linked to company c on day d. If a company has no scored article on that day, its sentiment contribution is set to zero for equal-weight market aggregation.

For the active EGX30 constituent set C_d, Equation 4.3 gives the daily market-mood feature:

$$M_d = \frac{1}{|C_d|} \sum_{c \in C_d} S_{c,d}. \tag{4.3}$$

This equal-weight design prevents heavily covered companies from dominating the index-level signal. Coverage counts, language counts, and constituent coverage ratios are retained as sentiment-side features because they describe the availability of the news signal.

All daily sentiment features are shifted by at least one calendar day before alignment to trading dates. News observed on calendar day d can therefore affect a prediction no earlier than the next available trading session.

4.6 EGX30 Target Construction

EGX30 closes are used only for label construction. For trading day t and horizon $h \in \{1, 3, 5\}$, Equation 4.4 defines the future return:

$$r_{t,h} = \frac{P_{t+h}}{P_t} - 1, \tag{4.4}$$

where P_t is the EGX30 close on trading day t. Equation 4.5 gives the binary direction label:

$$y_{t,h} = \mathbf{1}[r_{t,h} > 0]. \tag{4.5}$$

The close values, returns, and horizon-end dates are excluded from the feature matrix. This keeps the model sentiment-only and avoids converting the thesis into a price-based forecasting study.

4.7 Baselines and Model Evaluation

The primary direction model is logistic regression with standardized sentiment features. Ridge regression is retained only as a secondary return-oriented diagnostic in the repository; the thesis interpretation focuses on direction classification.

The direction model is compared with always-Up, always-Down, training-majority, rolling-majority, article-count-only, and shuffled-sentiment baselines. The always-class baselines expose label skew. The majority baselines test whether the model does more than follow historical class balance. The article-count-only baseline tests whether news volume alone explains performance. The shuffled-sentiment sanity check breaks sentiment timing while preserving the evaluation path.

The main metrics are accuracy, balanced accuracy, F1, MCC, actual Up share, predicted Up share, and delta versus the training-majority baseline. Balanced accuracy is prioritized because it is less likely than raw accuracy to reward majority-class prediction.

4.8 Supplementary Historical XGBoost Integration

After the sentiment-only benchmark is established, a supplementary integration experiment tests whether the same lagged market-mood signal improves a historical EGX30 forecasting model. This experiment uses the public EGX-Direction-Forecasting implementation by `anastamerr` as the historical-model reference and evaluates only its XGBoost classifier for the thesis discussion.

The historical-only variant uses Mubasher EGX30 Open, High, Low, Close, Volume (OHLCV) data from 21 April 2016 through 20 April 2026. The external preprocessor builds price-derived historical features from this series. The augmented variant keeps the identical historical feature matrix and appends 49 one-day-lagged and rolling bilingual market-mood sentiment features from the thesis pipeline. Both variants therefore share the same target definition, train/test split, scaling path, and test labels.

This supplementary experiment is intentionally separated from the main sentiment-only evaluation. Its target follows the historical-model repository: a positive class is assigned when the future 5-session average price is above the past 10-session average price. Rows whose future-window target cannot be observed are removed before scoring. The split is chronological, with a gap of 15 rows between train and test to cover the future window and return lags.

4.9 Leakage-Safe Validation and Robustness Checks

Panels are sorted chronologically and split into training, validation, and test regions with an embargo at least as long as the forecast horizon. Hyperparameters and classification thresholds are selected on validation predictions only. Validation and test predictions use expanding walk-forward histories limited to rows whose target labels would already be observable.

The validity audit recomputes targets from pinned EGX30 closes, verifies one-day lagging, checks active membership, checks cross-company deduplication, enforces walk-forward history limits, and recomputes saved metrics from predictions. Additional robustness reports summarize coverage concentration, class imbalance, baseline comparisons, abstention on high-signal days, sector aggregation scaffolding, and the unavailability of true historical index weights.

4.10 Chapter Summary

This chapter described the thesis method as a single reproducible workflow. The key design choice is to test bilingual financial-news sentiment without allowing price-derived predictors into the model matrix. Point-in-time membership, lagged aggregation, walk-forward evaluation, and validity audits are built into the design rather than added after the fact. The next chapter reports the empirical results.

Chapter 5

Results and Discussion

This chapter reports the empirical results of the final EGX30 sentiment-only study and a supplementary XGBoost price-plus-sentiment integration. It starts with the environment and dataset, then presents baselines, main sentiment results, the XGBoost integration result, robustness checks, the validity audit, and interpretation.

5.1 Experimental Environment

The final scripts were executed on a local macOS environment. Table 5.1 records the main reproducibility settings. Large generated artifacts are kept in repository output folders, while the thesis reports only the compact metrics needed for interpretation.

Table 5.1: Experimental environment for the final reproducible run.

Item	Value
Operating system	macOS 26.5 (Build 25F71), arm64
Python	3.9.6
Pandas	2.3.3
NumPy	1.26.4
Scikit-learn	1.6.1
Main direction model	Logistic regression on sentiment-only features
Supplementary hybrid model	XGBoost on EGX30 OHLCV plus lagged sentiment
Forecast horizons	1, 3, and 5 trading days
Validation design	Embargoed expanding-window walk-forward validation

5.2 Dataset and Coverage Summary

The final study covers 21 April 2021 through 20 April 2026. The point-in-time (PIT) registry contains 12 continuous membership intervals and 57 historical company identities. The base scored corpus contains 9,635 articles, with 7,418 Arabic articles and 2,217 English articles.

Table 5.2: Dataset and coverage summary.

Quantity	Value
Study period	21 April 2021–20 April 2026
Point-in-time membership intervals	12
Historical company identities	57
Scored articles before PIT scoping	9,635
Arabic scored articles	7,418
English scored articles	2,217
PIT-scoped market articles	4,782
Top 5 company article share	0.300
Top 10 company article share	0.486
Article-count HHI	0.037

After point-in-time membership filtering and cross-company deduplication, the primary scoped corpus contains 4,782 market articles. Table 5.2 summarizes the main dataset quantities used in the final results.

The Herfindahl-Hirschman Index (HHI) in Table 5.2 summarizes article concentration across companies. The coverage audit does not flag extreme concentration under the pre-set thresholds. The top five companies still account for about 30.0% of articles, so coverage is not uniform. This matters because sentiment-only forecasting is weaker on days when few constituents are represented.

5.3 Model Settings and Threshold Selection

The direction models use standardized sentiment-side features and logistic regression with balanced class weights. The regularization parameter C is selected from $\{0.01, 0.05, 0.1, 0.5, 1.0\}$ using validation walk-forward predictions only, with F1, balanced accuracy, and raw accuracy used as the selection order. This validation selection order follows the saved implementation, while the thesis interpretation prioritizes balanced accuracy because the test labels are imbalanced. Classification thresholds are also chosen from validation predictions only, using thresholds from 0.300 to 0.700 in steps of 0.025 and prioritizing balanced accuracy, then F1 and raw accuracy.

No test labels are used for hyperparameter or threshold selection. The selected C values and classification thresholds are stored with the saved metrics in the following repository files:

```
data/egx30_5yr_universe_metrics.csv
outputs/egx30_5yr_universe_experiments/metrics/direction_metrics.csv
```

Return-oriented ridge outputs are retained as repository diagnostics, but the thesis interpretation is based on direction classification.

5.4 Baseline Results

The baseline comparison tests whether the sentiment model adds information beyond simple class behavior or news volume. Table 5.3 reports average test metrics by baseline family.

Table 5.3: Average test performance by baseline family.

Baseline	Accuracy	Balanced accuracy	F1	MCC
Always-Up	0.666	0.500	0.799	0.000
Always-Down	0.334	0.500	0.000	0.000
Training-majority	0.666	0.500	0.799	0.000
Rolling-majority	0.666	0.500	0.799	0.000
Article-count-only logit	0.546	0.514	0.640	0.029
Shuffled-sentiment logit	0.519	0.506	0.598	0.012
Sentiment model	0.498	0.505	0.550	0.017

Each value is averaged across the available test horizons, language views, and sentiment-source routes in the saved baseline comparison report.

The always-Up and training-majority baselines have high average raw accuracy because the test labels lean upward. Their balanced accuracy remains exactly 0.500. This confirms why raw accuracy is not a sufficient headline metric for the thesis.

Table 5.3 is a compact summary over the saved baseline comparison rows. It is used to show the general class-imbalance pattern, not to replace the per-horizon baseline report. The full per-horizon and per-route baseline values are stored in `reports/egx30_baseline_comparison.csv`.

Figure 5.1 focuses on the clearest 1-day bilingual comparison. It shows that the best sentiment route improves over the majority baseline, but the lift is modest.

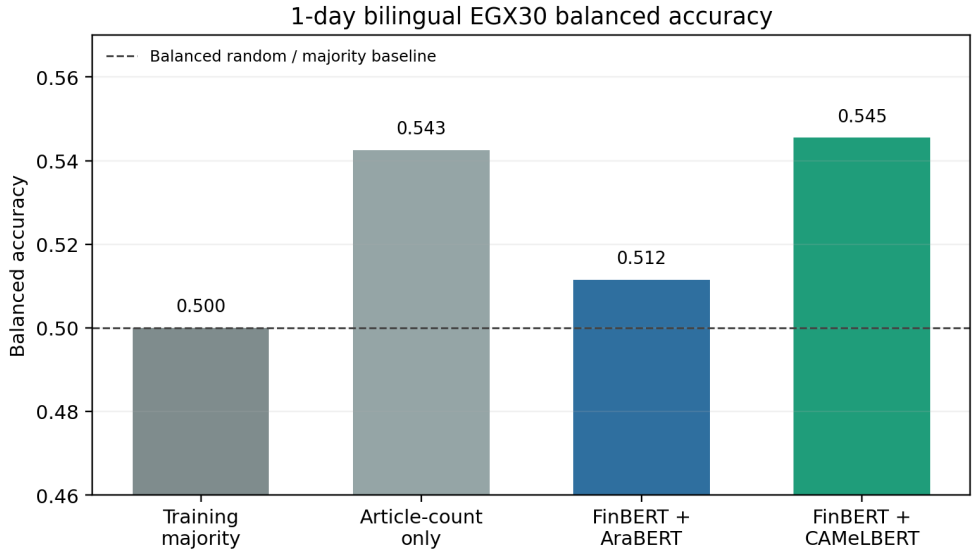


Figure 5.1: Balanced accuracy comparison for the 1-day bilingual EGX30 test rows.

5.5 Main Sentiment Forecasting Results

Table 5.4 reports the main bilingual test rows. The table includes class-distribution diagnostics so that each result can be interpreted alongside label skew.

Table 5.4: EGX30 bilingual sentiment-only test metrics under strict walk-forward evaluation.

Sentiment route	Horizon	Accuracy	Bal. acc.	F1	MCC	Actual Up	Pred. Up
FinBERT + AraBERT	1	0.522	0.512	0.591	0.023	0.621	0.547
FinBERT + AraBERT	3	0.404	0.416	0.447	-0.162	0.645	0.433
FinBERT + AraBERT	5	0.639	0.536	0.754	0.072	0.733	0.738
FinBERT + CAMeLBERT	1	0.621	0.545	0.737	0.115	0.621	0.823
FinBERT + CAMeLBERT	3	0.522	0.495	0.614	-0.009	0.645	0.591
FinBERT + CAMeLBERT	5	0.381	0.489	0.378	-0.021	0.733	0.262

The strongest bilingual row by balanced accuracy is FinBERT + CAMeLBERT at the 1-trading-day horizon, with accuracy 0.621, balanced accuracy 0.545, F1 0.737, and MCC 0.115. The predicted Up share is 0.823, so the raw accuracy is partly explained by upward-class behavior. The result is useful, but it is not strong enough to support a trading claim.

5.6 Supplementary Historical XGBoost Integration

The sentiment-only results answer whether text alone carries directional information. A second question is whether the same sentiment signal adds value when it is appended to a historical EGX30 forecasting model. Table 5.5 reports this supplementary XGBoost experiment.

The historical-only variant uses EGX30 OHLCV-derived features from the historical forecasting pipeline. The augmented variant adds 49 lagged and rolling bilingual market-mood sentiment features. Both rows use the same chronological split, the same 425 test rows, and the same target labels. The test period is 16 May 2024 through 8 April 2026.

Table 5.5: Supplementary XGBoost historical model with and without sentiment features.

Variant	Features	Accuracy	Bal. acc.	F1	AUC
Historical OHLCV only	18	0.739	0.757	0.789	0.848
Historical OHLCV + sentiment	67	0.802	0.793	0.851	0.839
Delta	+49	+0.064	+0.036	+0.062	-0.008

This is the clearest successful price-plus-sentiment result in the thesis artifacts. Adding sentiment to the historical XGBoost model improves raw accuracy by 6.4 percentage points, balanced accuracy by 3.6 percentage points, and F1 by 6.2 percentage points. AUC falls slightly, from 0.848 to 0.839, so the gain should be interpreted as an improvement in thresholded classification rather than a universal improvement in probability ranking.

The result does not replace the sentiment-only benchmark. Instead, it supports the practical interpretation that sentiment is more useful as an incremental feature in a historical model than as a standalone EGX30 direction signal.

5.7 Robustness and Sanity Checks

The robustness reports support the main interpretation rather than creating a separate research direction. Only 8 of 18 sentiment-model test rows beat the training-majority baseline on balanced accuracy. This prevents the thesis from claiming that sentiment consistently improves classification.

The abstention experiment asks whether predictions are more useful on high-signal days. The best row reaches balanced accuracy 0.580 at 35.6% test coverage, with 72 covered test samples. This is a conditional finding: it suggests that stronger sentiment days may be more informative, but it does not replace full-session evaluation.

Sector-aware aggregation is implemented as a compact scaffold using a manually reviewable sector map that tolerates `UNKNOWN`. True historical EGX30 constituent weights are not available in the reproducible dataset, so index-weighted sentiment is not reported. No market-cap, price, return, or Open, High, Low, Close, Volume (OHLCV) proxy is used as a substitute.

5.8 Validity Audit

The independent validity audit status is `PASS`. It checks 18 panels, 18,666 target rows, 18,666 lag rows, 7,290 validation and test predictions, and 36 saved metric rows. Table 5.6 summarizes the audit scope.

Table 5.6: Validity and robustness audit summary.

Check	Result
Point-in-time registry continuity	PASS
Inactive-company article exclusion	PASS
Cross-company article deduplication	PASS
EGX30 closes used for labels only	PASS
Daily sentiment shifted before trading-date alignment	PASS
Walk-forward history excludes unavailable labels	PASS
Saved metric parity against predictions	PASS
Easy-win report schema checks	PASS

These checks do not establish that sentiment is strongly predictive. They support a narrower claim: the reported results are produced by a leakage-safe and auditable protocol.

5.9 Interpretation and Limitations

The results are useful, but they are not strong enough to support a trading claim. Balanced accuracy is close to 0.500 for several rows, and the best bilingual row reaches 0.545. This is above a random balanced baseline but far from strong directional discrimination.

The baseline results explain why this matters. Always-Up and majority baselines can look strong under raw accuracy because the sample has an upward class bias. A thesis that selected results by raw accuracy would overstate the evidence. Selecting and interpreting results through balanced accuracy gives a more conservative conclusion.

Coverage also limits interpretation. The corpus is broad enough to support an index-level study, but some companies and sectors receive more news than others. Historical index weights are unavailable, so the thesis uses equal-weight market mood rather than claiming true index-weighted sentiment.

Bilingual financial-news sentiment from point-in-time EGX30 constituents appears to contain weak short-horizon directional information in some settings. The main value of the sentiment-only work is that it shows how to test that signal without look-ahead leakage or price-derived features. The supplementary XGBoost result then shows that the same signal can be more useful when added to a historical OHLCV model.

5.10 Chapter Summary

This chapter reported the final dataset, baselines, sentiment-only results, the supplementary XGBoost integration, robustness checks, and validity audit. The strongest standalone bilingual result is promising but limited, while the XGBoost integration provides a clearer successful incremental result. The validity audit supports the correctness of the evaluation protocol, and the metrics support a cautious claim: sentiment is weak alone but useful as an added historical-model feature. The next chapter concludes the thesis and identifies focused future work.

Chapter 6

Conclusion & Future Work

This final chapter summarizes what the thesis built, what the results show, and what should be done next. The conclusion remains intentionally conservative because the strongest results indicate weak directional signal, not strong market predictability.

6.1 Conclusion

This thesis built a reproducible point-in-time study of bilingual financial-news sentiment for EGX30 direction classification. The study reconstructs historical EGX30 membership, collects Arabic and English constituent-linked news, scores sentiment with language-specific encoders, aggregates daily market mood, constructs EGX30 direction labels, and evaluates the result with walk-forward validation.

The main empirical finding is conservative. The best bilingual row, FinBERT + CAMeLBERT at the 1-trading-day horizon, reaches accuracy 0.621 and balanced accuracy 0.545. This suggests limited but non-zero directional evidence. It does not show strong market predictability, and it does not justify a trading-system claim.

A supplementary price-plus-sentiment experiment gives a stronger applied result. When the lagged bilingual sentiment features are added to an EGX30 historical XGBoost model, test accuracy improves from 0.739 to 0.802, balanced accuracy improves from 0.757 to 0.793, and F1 improves from 0.789 to 0.851. This supports the practical conclusion that sentiment is more effective as an incremental feature in a historical model than as a standalone index-direction signal.

The baseline and class-imbalance checks are central to that conclusion. Always-Up and majority baselines can achieve attractive raw accuracy because the test labels lean upward. Balanced accuracy, MCC, predicted Up share, and baseline deltas prevent the thesis from overstating the model.

The validity audit supports the evaluation protocol. It verifies target construction, lag alignment, point-in-time membership, cross-company deduplication, walk-forward history, and metric parity. The contribution is therefore best understood as a conservative evaluation framework for bilingual EGX30 market mood.

6.2 Limitations

The study uses Mubasher as its main news source, so coverage may miss information from disclosures, broker reports, or other media. Coverage is also uneven across constituents even after point-in-time scoping. Some companies receive substantially more articles than others.

The sentiment labels come from off-the-shelf encoders rather than expert-labeled Egyptian financial sentiment data. FinBERT is finance-specific for English, but the Arabic route depends on general Arabic encoders. This limits how confidently the thesis can interpret Arabic sentiment quality.

The main model is intentionally sentiment-only. This isolates the text signal, but it also means the main benchmark should not be compared directly with full trading models that use prices, technical indicators, or macro variables. The supplementary XGBoost integration addresses this comparison only for one historical-model family and one target definition.

6.3 Future Work

Future work should start with expert-labeled Arabic financial sentiment data. Such a dataset would allow direct supervised evaluation of Arabic encoders in the financial domain instead of relying only on general Arabic sentiment models.

A second direction is richer event and materiality annotation. The current thesis uses sentiment and coverage signals, but it does not attempt complex event extraction. Future work could distinguish earnings, dividends, financing, regulation, governance, and contracts under a separately validated annotation scheme.

Broader auditable sources would also strengthen the protocol. Official EGX disclosures, company filings, and carefully archived news providers could improve coverage. True historical index weights should be added only if they become available from a reproducible source.

Finally, the successful XGBoost integration should be expanded into a broader price-only, sentiment-only, and price-plus-sentiment comparison under one unified point-in-time and walk-forward protocol. Social-media data should be considered only through approved access, careful entity filtering, and clear research-data handling rules.

Overall, the thesis shows that EGX30 sentiment forecasting should be judged through careful validation rather than raw accuracy alone. The measured signal is weak, but the evaluation setup is reusable and auditable. This makes the work a foundation for future Egyptian-market sentiment studies rather than a final trading model.

Appendix

Appendix A

Lists

List of Abbreviations

ANOVA Analysis of Variance.

API Application Programming Interface.

AraBERT Arabic Bidirectional Encoder Representations from Transformers.

ARBERT Arabic Bidirectional Encoder Representations from Transformers (Abdul-Mageed variant).

ARLUE Arabic Language Understanding Evaluation benchmark.

AZFinText Arizona Financial Text system.

BERT Bidirectional Encoder Representations from Transformers.

CAMeLBERT Computational Approaches to Modeling Language Bidirectional Encoder Representations from Transformers.

CIB Commercial International Bank.

CNN Convolutional Neural Network.

COMI Commercial International Bank (alternative EGX ticker symbol).

CPU Central Processing Unit.

CSV Comma-Separated Values.

DAE Denoising Autoencoder.

DJIA Dow Jones Industrial Average.

ECNN Emotion-enhanced Convolutional Neural Network.

EGX Egyptian Exchange.

EGX30 Egyptian Exchange 30-stock benchmark index.

FinBERT Financial Bidirectional Encoder Representations from Transformers.

GPOMS Google Profile of Mood States.

GPU Graphics Processing Unit.

GRU Gated Recurrent Unit.

GUC German University in Cairo.

HHI Herfindahl-Hirschman Index.

LLM Large Language Model.

LSTM Long Short-Term Memory.

MAE Mean Absolute Error.

MAPE Mean Absolute Percentage Error.

MARBERT Multi-dialect Arabic Bidirectional Encoder Representations from Transformers.

MCC Matthews Correlation Coefficient.

MLM Masked Language Modeling.

MLP Multilayer Perceptron.

NARX Nonlinear Autoregressive Exogenous model.

NLP Natural Language Processing.

OHLCV Open, High, Low, Close, Volume.

OIH Orascom Investment Holding.

PIT Point-in-Time.

RMSE Root Mean Squared Error.

RQ Research Question.

SA-DLSTM Sentiment-Aware Deep Long Short-Term Memory model.

SAHM Arabic Financial and Shari'ah-Compliant Reasoning benchmark.

SOFNN Self-Organizing Fuzzy Neural Network.

SVM Support Vector Machine.

TMG Talaat Moustafa Group.

TMGH Talaat Moustafa Group (alternative EGX ticker symbol).

VRAM Video Random-Access Memory.

XLM-R Cross-lingual Language Model RoBERTa.

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Appendix B

Reproducibility Notes

The main runner is `scripts/run_egx30_5yr_pipeline.py`. It rebuilds the point-in-time registry, prepares processed news, runs the EGX30 sentiment-only evaluation, executes validity checks, generates research summaries, checks thesis artifacts, and compiles the thesis.

The main evaluation and report scripts are listed below:

`scripts/run_egx30_5yr_evaluation.py`
`scripts/run_egx30_5yr_validity_checks.py`
`scripts/run_easy_win_validity_extensions.py`
`scripts/write_egx30_5yr_research_summary.py`

The compact reports are written under `reports/`.

The central reproducibility command is:

```
PYTHONPATH=src python3 scripts/run_egx30_5yr_pipeline.py --skip-scrape
```

The `--skip-scrape` flag reuses the local scrape cache. Removing it resumes the archival scraper, but no new scraping is required to reproduce the reported thesis results from the existing prepared artifacts.

Repository Report Pointers

The full baseline comparison is stored in:

`reports/egx30_baseline_comparison.csv`

The coverage audit, abstention experiment, sector scaffold, and weighting availability report are stored in:

`reports/egx30_coverage_audit.md`
`reports/egx30_abstention_experiment.md`
`reports/egx30_sector_feature_ablation.md`
`reports/egx30_weighting_ablation.md`

The main validity report, direction metrics, and selected model settings are stored in:

`data/egx30_5yr_validity_report.json`
`data/egx30_5yr_universe_metrics.csv`
`outputs/egx30_5yr_universe_experiments/metrics/direction_metrics.csv`

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